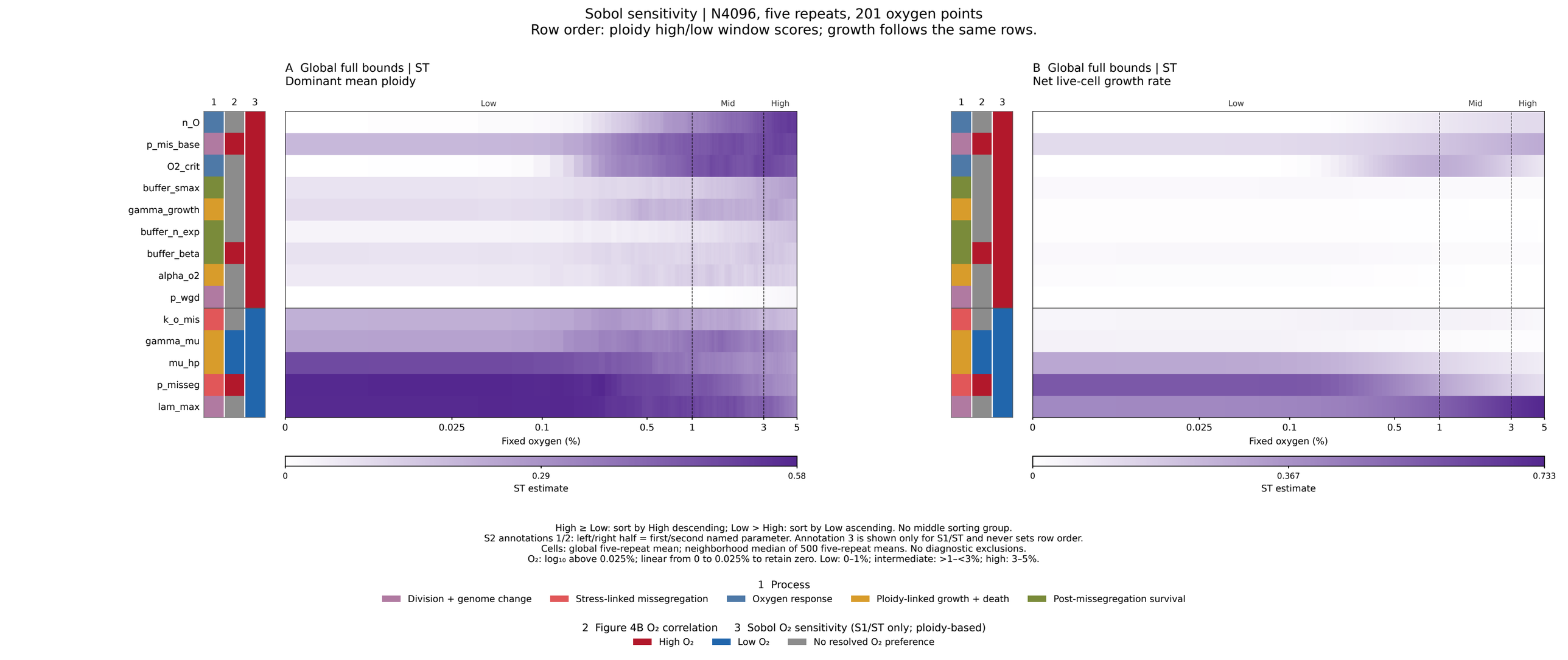
Sobol sensitivity | N4096, five repeats, 201 oxygen points
Row order: ploidy high/low window scores; growth follows the same rows.
A Global full bounds | ST
Dominant mean ploidy
1
2
3
n_O
p_mis_base
O2_crit
buffer_smax
gamma_growth
buffer_n_exp
buffer_beta
alpha_o2
p_wgd
k_o_mis
gamma_mu
mu_hp
p_misseg
lam_max
Low
Mid
High
0
0.025
0.1
0.5
1
3
5
Fixed oxygen (%)
0
0.29
0.58
ST estimate
B Global full bounds | ST
Net live-cell growth rate
1
2
3
Low
Mid
High
0
0.025
0.1
0.5
1
3
5
Fixed oxygen (%)
0
0.367
0.733
ST estimate
High ≥ Low: sort by High descending; Low > High: sort by Low ascending. No middle sorting group.
S2 annotations 1/2: left/right half = first/second named parameter. Annotation 3 is shown only for S1/ST and never sets row order.
Cells: global five-repeat mean; neighborhood median of 500 five-repeat means. No diagnostic exclusions.
O₂: log₁₀ above 0.025%; linear from 0 to 0.025% to retain zero. Low: 0–1%; intermediate: >1–<3%; high: 3–5%.
1 Process
Division + genome change
Stress-linked missegregation
Oxygen response
Ploidy-linked growth + death
Post-missegregation survival
2 Figure 4B O₂ correlation
3 Sobol O₂ sensitivity (S1/ST only; ploidy-based)
High O₂
Low O₂
No resolved O₂ preference